05 · Data Binding

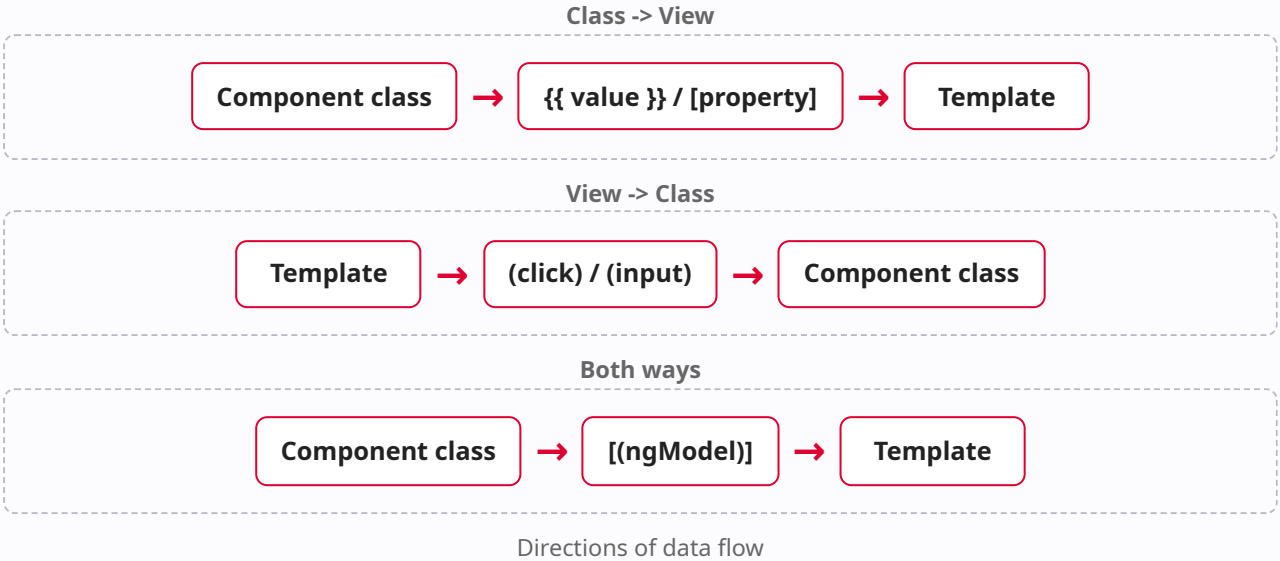
Beginner Angular learning notes

a. Definition

Data binding connects the component class to its template so the screen stays in sync with your data. Angular has four kinds: **interpolation** `{{ }}` (show a value), **property binding** `[prop]` (set an element property), **event binding** `(event)` (react to user actions) and **two-way binding** `[(ngModel)]` (both directions at once).

Remember the brackets: `[ ]` sends data *into* the template, `( )` sends events *out* of it, and `[( )]` (banana in a box) does both. For `ngModel` you must import `FormsModule`.

b. Diagram



Directions of data flow

c. Syntax

```
{{ expression }}                <!-- interpolation -->
<button [disabled]="isBusy">      <!-- property binding -->
<div [attr.aria-label]="label">   <!-- attribute binding -->
<button (click)="save()">         <!-- event binding -->
<input [(ngModel)]="name" />      <!-- two-way (needs FormsModule) -->
```

d. Example Program

`Programs/05_Data_Binding/data-binding.component.ts`

```
import { Component } from '@angular/core';
import { FormsModule } from '@angular/forms'; // needed for [(ngModel)]

@Component({
  selector: 'app-root',
  imports: [FormsModule],
```

```
  template: `
    <!-- interpolation: show a class value as text -->
    <h2>{{ brand }}</h2>

    <!-- two-way binding [(ngModel)] + attribute binding [attr.aria-label] -->
    <input [(ngModel)]="name" placeholder="Your name" [attr.aria-label]="'Name for ' + brand"
/>

    <!-- event binding (click) + property binding [disabled] -->
    <button (click)="greet()" [disabled]="name.trim() === ''">Greet</button>

    <p>You typed: {{ name }}</p>
    <p>{{ greeting }}</p>
    <p>Button clicked {{ clicks }} time(s)</p>
  `,
})
export class DataBindingComponent {
  brand = 'Chennai Tea House';
  name = '';
  greeting = '';
  clicks = 0;

  greet() {
    this.greeting = `Vanakkam, ${this.name}!`;
    this.clicks++;
  }
}
```

e. Output

Rendered in the browser (text on screen, after the UI steps below):

1. Type `Karthik` in the text box.
2. Click the **Greet** button twice.

```
Chennai Tea House
Greet
You typed: Karthik
Vanakkam, Karthik!
Button clicked 2 time(s)
```

f. Real-World Example in Tamil (தமிழ் உதாரணம்)

தேநீர்க் கடையில் நீங்கள் 'ஒரு டீ, சர்க்கரை குறைவா' என்று சொல்கிறீர்கள். மாஸ்டர் உடனே அதை கவனித்து அதன்படி டீ போடுகிறார்; விலைப்பட்டியல் பலகையில் உள்ள விலையை நீங்கள் படித்துக்கொள்கிறீர்கள். இரண்டு பக்கமும் தகவல் பரிமாறப்படுகிறது.

பலகையில் உள்ள விலை = `{{ price }}` (class-இலிருந்து திரைக்கு). நீங்கள் சொல்லும் ஆர்டர் = `(click)` அல்லது `(input)` (திரையிலிருந்து class-க்கு). இரண்டும் எப்போதும் ஒத்திசைவாக இருப்பது = `[(ngModel)]`.

g. Common Mistakes

- Forgetting to import `FormsModule`, which causes "Can't bind to 'ngModel'".

- Writing `{{ }}` inside a property binding like `[title]="{{ x }}"`; use `[title]="x"` instead.
- Mixing up `[ ]` and `( )` brackets.
- Putting heavy logic or function calls inside templates; they run on every change detection.

h. Interview Questions

Q1. What are the types of data binding in Angular?
Interpolation, property binding, attribute binding, class/style binding, event binding and two-way binding.

Q2. Interpolation vs property binding?
Interpolation converts the value to a string inside text; property binding sets a DOM property with its real type (boolean, object…).

Q3. What is two-way binding?
Combining property and event binding, written `[(ngModel)]`, so the view and the class update each other.

Q4. When do you use `[attr.x]`?
When there is no DOM property for it (like `aria-*` or `colspan`), so you must set the HTML attribute.

i. Practice Exercises

1. Build a temperature converter: an input for Celsius and text showing Fahrenheit.
2. Add a button that is disabled until a checkbox is ticked.
3. Make a character counter that shows how many letters are typed in a textarea.